DSP – LEC – 6 - Ch-2

CHAPTER – 2

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DISCRETE TRANSFORMS (DFT & FFT)

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Sec. 2.4

2.4 COMPUTATIONAL COMPLEXITY OF THE DFT

A large number of multiplications and additions are required for the calculation of the DFT. For an 8-point DFT the expansion for X(k) becomes (from Eq. 2.4).

$$X(k)=\sum_{n=0}^{7} x(n)e^{-\frac{2\pi jkn}{8}}, \qquad k=0,\ldots\ldots,7$$

and letting $\frac{k2\pi}{8} = K$ **this expands to:**

$$X(k) = x(0)e^{-jk0} + x(1)e^{-jk1} + x(2)e^{-jk2} + x(3)e^{-jk3} + x(4)e^{-jk4}+x(5)^{-jk5}+x(6)e^{-jk6}+ x(7)e^{-jk7},$$

$$k=0,1, ..7 \qquad (2.15)$$

Sec. 2.4 Cont. - 1

$$X(k) = x(0)e^{-jk0} + x(1)e^{-jk1} + x(2)e^{-jk2} + x(3)e^{-jk3} + x(4)e^{-jk4} + x(5)^{-jk5} + x(6)e^{-jk6} + x(7)e^{-jk7},$$

$k=0,1, ..7$ (2.15)

Each of the product terms is added together. There are therefore eight complex multiplications and seven complex additions to be calculated. For N-point DFT, there will be N and N - 1 of them respectively.

There are also eight harmonic components to be evaluated (k = 0, . . ., 7). This number becomes N for an N -point DFT.

So, for an N-point DFT these complex multiplications and complex additions to be calculated become N^2 and N(N-1) respectively.

Sec. 2.4 Cont. - 2

Numbers complex multiplications and complex additions in DFT is summarized in the following table:

Operation	DFT
No. of Complex Addition	N (N-1)
No. of Complex Multiplications	N^2

If N = 1024, then approximately one million complex multiplications and one million complex additions are required.

Clearly some means of reducing these numbers is desirable.

For example,

if k =1 and n=2, so,

$$e^{\frac{-2\pi jkn}{8}} = e^{\frac{-j\pi}{2}}, \text{ and}$$

if k =2 and n =1,

$$e^{\frac{-2\pi jkn}{8}} = e^{\frac{-j\pi}{2}}, \text{ also}$$

Sec. 2.5

2.5 DECIMATION-in- TIME FFT ALGORITHM

In this section it will be shown how the computational redundancy inherent in the DFT is used to reduce the number of different calculations necessary, thereby speeding up the computation.

For a 1024-point DFT the number of calculations required can be reduced by a factor of 204.8. The algorithms which can achieve this are given the title 'fast Fourier transform', or FFT in short.

When applied in the time domain the algorithm is referred to as a decimation-in-time (DIT) FFT. The first DIT algorithm was due to Cooley and Tukey (1965), after whom it is often named. Decimation then refers to the significant reduction in the number of calculations performed on time domain data. It is noteworthy that the computational savings will be seen to increase as $N^2 - (N/2)\log_2 N$.

Sec. 2.5 Cont. - 1

First:

The notation will be simplified and some mathematical relationships will be established. Thus Eq. 2.4 will be re-written as

$$X(k)=\sum_{n=0}^{N-1} x(n)e^{-\frac{2\pi jkn}{8}}, \qquad k=0,\ldots..,N-1 \quad (2.16)$$

The factor $e^{\frac{-j2\pi}{N}}$ **will be written as** W_N**, thus**

$$W_N= e^{\frac{-j2\pi}{N}} \qquad (2.17)$$

So that Eq. 2.16 becomes

$$X(k)=\sum_{n=0}^{n} X_n W_N^{kn}, \qquad k=0,\ldots\ldots..,N-1 \quad (2.18)$$

It is worthwhile at this point to note some relationships involving W_N**.**

First,

$$W_N^2=\left(e^{\frac{-j2\pi}{N}}\right)^2 = e^{\frac{-j2*2\pi}{N}} = e^{\frac{-j2\pi}{N/2}} = W_{N/2} \qquad (2.19)$$

Second,

$$W_N^{(k+N/2)}= W_N^k W_N^{N/2} = W_N^k\, e^{-j\left(\frac{2\pi}{N}\right)\left(\frac{N}{2}\right)} = W_N^k\, e^{-j\pi} = -W_N^k \qquad (2.20)$$

Sec. 2.5 Cont. - 2

Summarizing the useful results concerning W_N for convenience, we have

$$W_N = e^{\frac{-j2\pi}{N}} \qquad (2.21a)$$

$$W_N^2 = W_{N/2} \qquad (2.21b)$$

$$W_N^{(k+N/2)} = -W_N^k \qquad (2.21c)$$

The suffixes, n, in Eq. 2.18 extend from n = 0 -to- n=N - 1, corresponding to the data values x_0, x_1, x_2, x_3, ..., x_{N-1}. The even-numbered sequence is x_0, x_2, x_4…, x_{N-2} and the odd-numbered sequence is x_1, x_3, . . ., x_{N-1}.

Both sequences contain N/2 points. The terms in the even sequence may be designated x_{2n} with n=0 -to- n=N/2 -1 while those in the odd sequence become x_{2n+1}.

$$X_1(k) = \underbrace{\sum_{0}^{\frac{N}{2}-1} x_{2n} W_N^{2nk}}_{even\,sequence} + \underbrace{\sum_{0}^{\frac{N}{2}-1} x_{2n+1} W_N^{(2n+1)k}}_{odd\,sequence} = \sum_{n=0}^{\frac{N}{2}-1} x_{2n} W_N^{2nk} + W_N^k \sum_{n=0}^{\frac{N}{2}-1} x_{2n+1} W_N^{2nk}$$

$$k = 0 , \ldots\ldots , N - 1 \qquad (2.22)$$

Sec. 2.5 Cont. -3

- Using Eq. 2.21b gives $W_N^{2nk} = W_{N/2}^{nk}$ so Eq. 2.22 becomes

$$\sum_{n=0}^{\frac{N}{2}-1} x_{2n} W_{N/2}^{nk} + W_N^k \sum_{n=0}^{\frac{N}{2}-1} x_{2n+1} W_{N/2}^{nk}, \qquad k=0, \ldots, N-1 \qquad (2.23)$$

Equation 2.23 may be written as:

$$X_1(k) = X_{11}(k) + W_N^k X_{12}(k), \qquad k=0, \ldots, N-1 \qquad (2.24)$$

On comparison of Eq. 2.24 with Eq. 2.18, it is seen that $X_{11}(k)$ is indeed the DFT of the even sequence, while $X_{12}(k)$ is that of the odd sequence. Therefore, as previously stated, the DFT, $X_1(k)$, can be expressed in terms of two DFTs: $X_{11}(k)$ and $X_{12}(k)$. The factor $W_{N/2}^k$ occurs in both $X_{11}(k)$ and $X_{12}(k)$ and needs calculation once only.

Sec. 2.5 Cont. -4

Table 2.1 illustrates the process for an 8-point DFT. Line 1 gives the data while line 2 gives an expression for the DFT of the data in terms of the DFTs of the even and odd sequences, $X_{11}(k)$ and $X_{12}(k)$ respectively. Line 3 shows the re-ordered data from which $X_{11}(k)$ and $X_{12}(k)$ are derived. Line 4 gives the DFTs of the data sequences of line 3 in terms of the DFTs of their even and odd sequences, $X_{21}(k)$, $X_{22}(k)$, $X_{23}(k)$ and $X_{24}(k)$. These sequences are shown in line 5, and are seen to be the ultimate 2-point sequences, whose DFTs are $X_{21}(k)$, $X_{22}(k)$, $X_{23}(k)$ and $X_{24}(k)$ and are expressed in terms of the data in line 6. Thus the single 8-point DFT has been decomposed into four 2-point DFTs, each one of which produces two values, for example $X_{21}(0)$ and $X_{21}(1)$ in the case of $X_{21}(k)$. This process involved two decompositions, and the weights W_N^k were squared at each step.

Sec. 2.5 Cont. -5

Table 2.1 Structure of an 8- Point FFT

Line number	*Line content*								*k ranges*	*N ranges*
1	Data sequence A_0	A_0	$x_0\ x_1\ x_2\ x_3\ x_4\ x_5\ x_6\ x_7$							$0, \ldots, 7$
2	8-point DFT of A_0	$X_1(k) = X_{11}(k) + W_N^k X_{12}(k)$							$0, \ldots, N-1$ $(0, \ldots, 7)$	$0, \ldots, 7$
3	Re-ordered A_0: two sequences, A_1 and A_2	A_1	$x_0\ x_2\ x_4\ x_6$			A_2	$x_1\ x_3\ x_5\ x_7$			$0, \ldots, 3$
4	4-point DFTs of A_1 and A_2	$X_{11}(k) = X_{21}(k) + W_{N/2}^k X_{22}(k)$				$X_{12}(k) = X_{23}(k) + W_{N/2}^k X_{24}(k)$			$0, \ldots, N/2-1$ $(0, \ldots, 3)$	$0, \ldots, 3$
5	Re-ordered sequences A_1 and A_2: four sequences A_3, A_4, A_5, A_6	A_3	$x_0\ x_4$	A_4	$x_2\ x_6$	A_5	$x_1\ x_5$	A_6 $x_3\ x_7$		$0, 1$
6	2-point DFTs of A_3, A_4, A_5, A_6	$X_{21}(k) = x_0 + W_{N/4}^k x_4$		$X_{22}(k) = x_2 + W_{N/4}^k x_6$		$X_{23}(k) = x_1 + W_{N/4}^k x_5$		$X_{24}(k) = x_3 + W_{N/4}^k x_7$	$0, \ldots, N/4-1$ $(0, 1)$	$0, 1$

Sec. 2.5 Cont. -6

Using table 2.1, considering line 6, we can reach to the following decomposing equations (2.25)

$$X_{11}(0) = X_{21}(0) + X_{22}(0) = X_{21}(0) + W_8^0 X_{22}(0) \quad (2.25a)$$

$$X_{11}(2) = X_{21}(0) - X_{22}(0) = X_{21}(0) - W_8^0 X_{22}(0) \quad (2.25b)$$

$$X_{11}(1) = X_{21}(1) - jX_{22}(1) = X_{21}(1) + W_8^2 X_{22}(1) \quad (2.25c)$$

$$X_{11}(3) = X_{21}(1) + jX_{22}(1) = X_{21}(1) - W_8^2 X_{22}(1) \quad (2.25d)$$

The number of complex additions and multiplications involved is reduced in this way because:

(i) the re-composition equations are expressed in terms of powers of the recurring factor W_N,

(ii) use is also made of relationships of the type $X_{21}(2) = X_{21}(0)$ and $X_{21}(3) = X_{21}(1)$, and

(iii) the presence of only sign differences in the pairs of expressions is exploited. This algorithm is known as the Cooley-Tukey algorithm.

Sec. 2.5.1

2.5.1 The butterfly

Equations 2.25 may be represented diagrammatically by exploiting the symmetry centered on the sign differences and taking the equations in pairs.

Thus from Eq.s. 2.25a and 2.25b the outputs of the re-compositions are $X_{11}(0)$ and $X_{11}(2)$ formed from the inputs $X_{21}(0)$ and $X_{22}(0)$. This is illustrated in Fig. 2.2(a).

The inputs are at the left-hand side of the cross, the outputs to the right. Figure 2.2(b) shows how the outputs $X_{11}(1)$ and $X_{11}(3)$ are obtain diagrammatically.

By overlapping Figs. 2.2(a) and 2.2(b), a composite diagram is obtained in which the output DFTs are arranged in order of increasing **k**. This is shown in Fig. 2.2(c).

The structure of Fig. 2.2(a) or 2.2(b) is referred to as a **'butterfly'**, being reminiscent of the symbolic representation of the insect.

The entire 8-point FFT may be depicted in this manner as Fig. 2.3.

Sec. 2.5.1 Cont. -1

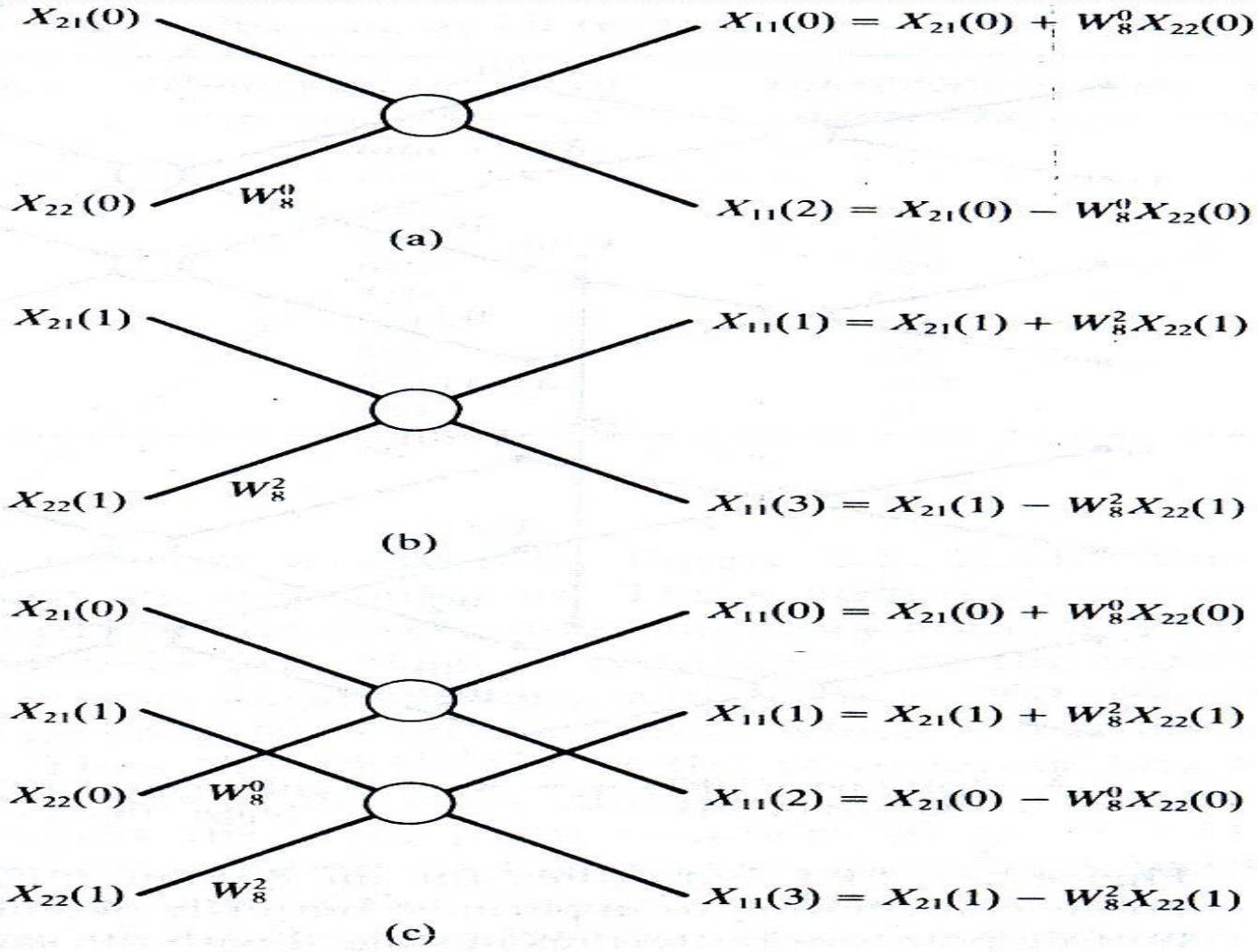


Fig. 2.2 FFT butterflies.

Sec. 2.5.1 Cont. -2

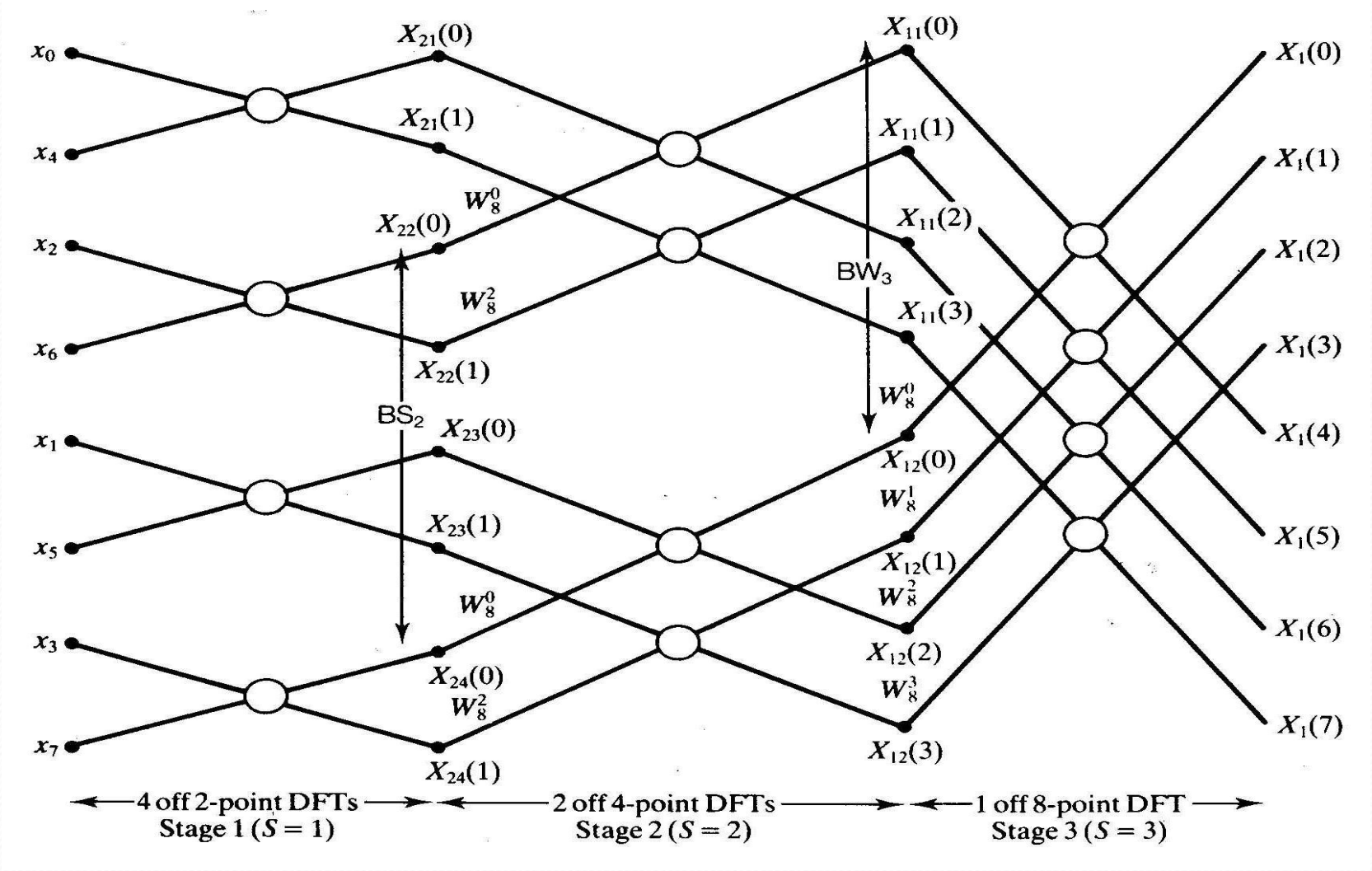


Fig. 2.3 DFT butterflies for an 8-point DFT.

Sec. 2.5.1 Cont. -3

Example 2.3

It will now be instructive to obtain the DFT of the sequence {1, 0, 0, 1}, previously evaluated in Section 2.1, by means of the decimation-in-time FFT algorithm. This is a four-point DFT with $x_0 = 1$, $x_1 = 0$, $x_2 = 0$, $x_3 = 1$, and

$$X_1(k) = X_{11}(k) + W_N^k X_{12}(k), \qquad k = 0, 1, 2, 3.$$

The re-ordered sequence is x_0, x_2, x_1, x_3.

We can now utilize the top left-hand corner of Fig. 2.3 to work out the DFT. Points x_0, x_4, x_2, x_6 are replaced by x_0, x_2, x_1, x_3, and the required DFT values are $X_{11}(0)$, $X_{11}(1)$, $X_{11}(2)$ and $X_{11}(3)$.

$$X_{21}(0) = x_0 + x_2 = 1 \qquad X_{21}(1) = x_0 - x_2 = 1$$

$$X_{22}(0) = x_1 + x_3 = 1 \qquad X_{22}(1) = x_1 - x_3 = -1$$

Sec. 2.5.1 Cont. -4

Therefore,

$X_{21}(0) = x_0 + x_2 = 1$ $\quad$ $X_{21}(1) = x_0 - x_2 = 1$

$X_{22}(0) = x_1 + x_3 = 1$ $\quad$ $X_{22}(1) = x_1 - x_3 = -1$

$X_{11}(0) = X_{21}(0) + W_8^0 X_{22}(0) = 1 + 1 = 2$

$X_{11}(3) = X_{21}(1) + W_8^2 X_{22}(1) = 1 + e^{-j\pi/2}(-1) = 1 + j$

$X_{11}(2) = X_{21}(0) - W_8^0 X_{22}(0) = 1 - 1 = 0$

$X_{11}(3) = X_{21}(1) - W_8^2 X_{22}(1) = 1 - j$

These values are the same as those obtained in Section 2.1, but were more readily obtained using the FFT algorithm. This conclusion is general and the computational savings increase as the number of data increases.

Sec. 2.5.2

2.5.2 Computational advantages of the FFT

The computational advantages of the FFT may be illustrated by considering first the FFT algorithm of Fig. 2.3.

This figure shows that an N-point FFT contains N/2 butterflies per stage and $\log_2$ N stages, that is it contains a total of (N/2) $\log_2$ N butterflies.

Figure 2.4(a) shows that each butterfly involves one complex multiplication of the form $W_N^R X_{ij}(k)$.

Hence the FFT involves (N/2) $\log_2$ N complex multiplications compared with N^2 in the case of the DFT, as shown in Section 2.4.

Thus the computational saving in complex multiplications is N^2 - (N/2) log2 N.

Sec. 2.5.2 Cont. -1

Each butterfly contains two complex additions so the FFT requires N $\log_2$ N complex additions compared with N (N - 1) for the DFT.

Thus the saving in complex additions is N(N - 1) - N $\log_2$ N.

These savings are illustrated in Table 2.2.

Table 2.2 Saving in complex multiplications and additions when the FFT is used instead of the DFT.

	DFT		*FFT*		*Ratio of DFT multiplications to FFT multiplications*	*Ratio of DFT additions to FFT additions*
N	*Number of complex multiplications*	*Number of complex additions*	*Number of complex multiplications*	*Number of complex additions*		
2	4	2	1	2	4	1
4	16	12	4	8	4	1.5
8	64	56	12	24	5.3	2.3
16	256	240	32	64	8.0	3.75
32	1 024	992	80	160	12.8	6.2
64	4 096	4 032	192	384	21.3	10.5
128	16 384	16 256	448	896	36.6	18.1
256	65 536	65 280	1 024	2 048	64.0	31.9
512	262 144	261 632	2 304	4 608	113.8	56.8
1024	1 048 576	1 047 552	5 120	10 240	204.8	102.3
2048	4 194 304	4 192 256	11 264	22 528	372.4	186.1
4096	16 777 216	16 773 120	24 576	49 152	682.7	341.3
8192	67 108 864	67 100 672	53 248	106 496	1 260.3	630.0

Sec. 2.6

2.6 INVERSE FFT ALGORITHM

To see how the IFFT is derived make the following substitutions in Eq. 2.4 (DFT).

Sum over the variable λ rather than n, let the variable k become μ, set $\Omega = 2\pi /NT$ so that the exponent of "e" becomes: -j μ (2π /N)λ.

Again using the notation x(λT) = x(λ) and so on. Equation 2.4 then becomes:

$$X(\mu) = \sum_{\lambda=0}^{N-1} x(\lambda) e^{-j\mu\,(2\pi/N)\lambda} \qquad \mu = 0, 1, \ldots, N-1 \qquad (2.26)$$

Equation 2.26, represent the FFT.

Sec. 2.6 Cont. -1

Now make similar substitutions in Eq. 2.7 (IDFT), that is put k=λ, and n=μ so that Eq. 2.7 becomes:

$$\mathbf{X}(\mu) = \frac{1}{N}\sum_{\lambda=0}^{N-1} x(\lambda) e^{j\mu\,(2\pi\,/\mathrm{N})\lambda} \qquad \mu = 0, 1, \ldots\ldots, N-1 \qquad (2.27)$$

Equation 2.27, represent the IFFT.

We can see that the IFFT, x(μ), (2.27) differs from the FFT, X(μ), (2.26) only in the factor 1/N and the sign of the exponent.

Thus with small modifications the FFT may be used to calculate the IFFT.

Sec. 2.6 Cont. -2

Example 2.4

The first four sampled voltage values of a 10Hz bandwidth signal sampled at 125Hz were (0.5, 1, 1, 0.5). Demonstrate how the discrete Fourier transform of this sequence may be obtained using the fast Fourier transform, and hence obtain the Fourier transform of the data.

Solution

The flowgraph for the FFT is given in Fig. 2.4. We have

$$X(0) = G(0) + W_4^0 H(0) \qquad = G(0) + H(0)$$

$$G(0) = x(0) + W_2^0 x(2) \qquad = x(0) + x(2)$$

$$H(0) = x(1) + W_2^0 x(3) \qquad = x(1) + x(3)$$

Sec. 2.6 Cont. -3

Since $W^0 = 1$, substituting values gives

$X(0) = x(0) + x(2) + x(1) + x(3)$

$= 0.5 + 1 + 1 + 0.5 \quad = 3$

$X(1) = G(1) + W_4^1 H(1)$

$G(1) = x(0) - W_2^0 x(2) \quad = x(0) - x(2)$

$H(1) = x(1) - W_2^0 x(3) \quad = x(1) - x(3)$

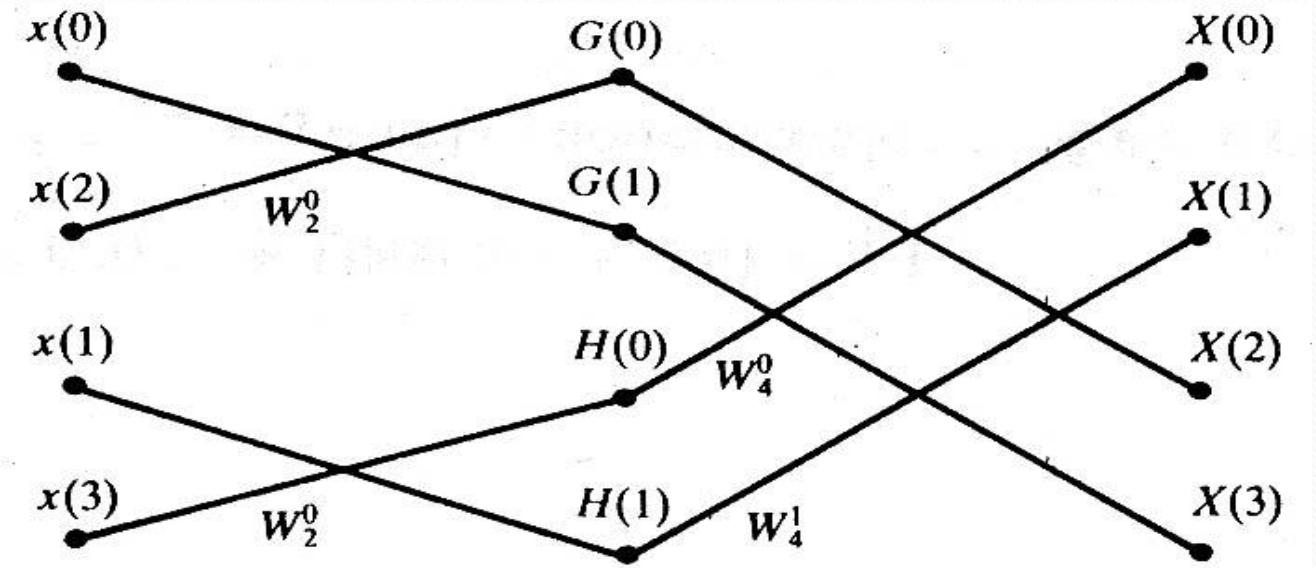


Fig. 2.4 Flowgraph of the FFT in Example 2.4.

Now, $W_N = e^{\frac{-j2\pi}{N}}$, , and therefore

$W_N^1 = = e^{-j2\pi/4} = e^{-j2\pi/2}$

Sec. 2.6 Cont. -4

Substituting gives

$$X(1) = x(0) - x(2) + e^{-j\pi/2}[x(1) - x(3)]$$

$$= 0.5 - 1 + \left[\cos\left(\frac{\pi}{2}\right) - j\sin\left(\frac{\pi}{2}\right)\right](1 - 0.5)$$

$$= 0.5 - 1 + (0 - j)0.5 = -0.5 - j0.5 = -0.5(1 + j)$$

$$X(2) = G(0) - W_4^0 H(0) \quad = G(0) - H(0)$$

$$= x(0) + x(2) - [x(1) + x(3)]$$

$$= 0.5 + 1 - (1 + 0.5) = 0$$

$$X(3) = G(1) - W_4^1 H(1)$$

$$= x(0) - x(2)\, e^{-j\pi/2}\, [x(1) - x(3)]$$

$$= 0.5 - 1 - \left[\cos\left(\frac{\pi}{2}\right) - j\sin\left(\frac{\pi}{2}\right)\right](1 - 0.5)$$

$$= -0.5 - (-j)0.5 = -0.5 + j0.5 = 0.5(-1 + j)$$

Therefore $X(\Omega) = \{3, -0.5(1 + j), 0, 0.5(-1 + j)\}$

Sec. 2.6 Cont. -5

When T, the sampling interval, is small,

$$FT = T\ DFT$$

Where FT is the Fourier transform.

Here, T = 1/125 s = 0.008 s.

The signal period is 1/10 s = 0.1 s; therefore: $\frac{T}{period} = \frac{0.008}{0.1} = 0.08 << 1$

and it is a good approximation to put FT = T DFT, so that:

$$FT = \{ 0.024, -0.004(1 + j), 0, 0.004(-1 + j)\}$$